PERFOMANCE TESTING

Performance Testing

Performance testing is carried out to evaluate the responsiveness, stability, reliability, and scalability of the **Medical Diagnosis Centre Management System** under different workloads. The testing focuses on the **Service Catalog, patient request submission, record creation, diagnostic test assignment, workflow execution, notifications, and Flow Designer processing** .

1. Performance Testing Objectives

- Verify system behavior under normal and peak workloads.
- Measure response time for major system operations.
- Verify that patient and diagnostic requests are processed successfully under concurrent usage.
- Identify performance bottlenecks.
- Monitor system resource utilization.
- Verify workflow execution performance.
- Evaluate the scalability of the solution.
- Ensure that performance does not negatively affect the user experience.

2. Performance Test Environment

Component	Configuration
Platform	ServiceNow
Application	Medical Diagnosis Centre Management System
Instance	Personal Developer Instance / Development Instance
Main Interface	Service Catalog
Workflow	Flow Designer
Database	ServiceNow Tables
Test Users	Multiple simulated/concurrent users
Test Data	Patient records, appointments, diagnostic test requests, approvals, and catalog variables
Network	Stable internet connection
Testing Type	Functional performance, load, stress, and concurrency testing

3. Performance Test Scenarios

ID	Test Scenario	Expected Result
PT01	Open Medical Diagnosis Request catalog item	Form loads successfully without noticeable delay
PT02	Submit a patient/diagnosis request	Request is submitted successfully
PT03	Submit request with all required variables	All required patient and diagnostic information is captured correctly
PT04	Create patient/diagnosis record	Record is created successfully
PT05	Execute diagnostic workflow	Workflow is triggered successfully

ID	Test Scenario	Expected Result
PT06	Assign diagnostic test	Test is assigned to the appropriate staff/process
PT07	Process diagnostic request	Diagnostic information/status is updated correctly
PT08	Send notification	Notification is generated successfully
PT09	Update diagnosis record	Record status is updated correctly
PT10	Execute complete Flow Designer workflow	End-to-end workflow completes successfully
PT11	Multiple users submit requests simultaneously	System processes concurrent requests correctly
PT12	Multiple diagnostic requests are processed	Workflow remains stable
PT13	Search and filter patient/diagnosis records	Results are returned correctly
PT14	Open diagnosis records repeatedly	Records load consistently
PT15	High-volume diagnosis request submission	System remains stable under increased workload

4. Performance Metrics

Metric	Description
Response Time	Time taken by the system to respond to a user action
Throughput	Number of diagnosis requests processed during a specific period
Concurrent Users	Number of users performing operations simultaneously
Error Rate	Percentage of failed transactions
Workflow Execution Time	Time required to complete the automated diagnosis workflow
Database Processing Time	Time required for patient, appointment, test, and report record operations
Notification Processing Time	Time required to generate and send notifications
CPU Utilization	Processor utilization during testing
Memory Utilization	Memory consumption during testing
SLA Performance	Ability of the system to process requests within defined SLA targets

5. Load Testing

Load testing evaluates the application under expected user workloads.

Load Level	Example Users	Purpose
Light Load	10	Verify basic system performance
Normal Load	25	Represent normal usage
Heavy Load	50	Evaluate increased workload
Peak Load	100	Evaluate peak concurrent usage

Load Level	Example Users	Purpose
Stress Load	150+	Identify system limits and performance degradation

These values are **test-planning examples** and should be adjusted according to the actual expected users of the Medical Diagnosis Centre system.

6. Concurrency Testing

Concurrency testing verifies whether multiple users can perform operations at the same time without data loss or workflow failure.

Examples:

- Multiple users submitting diagnosis requests simultaneously.
- Multiple patient requests entering the workflow.
- Multiple patient or appointment records being created simultaneously.
- Multiple diagnostic tests being processed simultaneously.
- Multiple notifications being triggered.
- Multiple staff members updating diagnosis records.

Expected Result

The system should process concurrent transactions correctly without duplicate records, lost information, workflow failures, or incorrect status updates.

7. Stress Testing

Stress testing gradually increases the workload beyond normal operating conditions to identify the point at which system performance begins to degrade.

The test should monitor:

- Response time
- Error rate
- Workflow execution
- Record creation
- Database operations
- Notification processing
- Resource utilization

The purpose is to identify system limitations and determine whether additional optimization or configuration is required.

8. End-to-End Performance Test

The complete diagnosis request lifecycle can be tested as one transaction:

Submit Patient Request → Capture Patient Details → Create Record → Assign Diagnostic Test → Process Diagnosis → Flow Logic → Notification → Update Record → Closure

The project workflow uses Flow Designer to automate the major stages of request processing.

9. Performance Test Case Template

Test Case ID	Operation	Input	Expected Result	Actual Result	Status
PT01	Open Diagnosis Request	Patient request form	Form opens successfully	To be measured	Pending
PT02	Submit Request	Valid patient details	Request submitted successfully	To be measured	Pending

				Actual Result	Status
PT03	Create Record	Submitted request	Patient/diagnosis record created	To be measured	Pending
PT04	Assign Diagnostic Test	Valid diagnosis request	Diagnostic test assigned successfully	To be measured	Pending
PT05	Process Diagnosis	Assigned diagnostic test	Diagnosis process continues successfully	To be measured	Pending
PT06	Update Diagnosis	Processed test details	Diagnosis record updated correctly	To be measured	Pending
PT07	Notification	Request/test status	Notification generated successfully	To be measured	Pending
PT08	Update Record	Completed diagnosis	Status updated correctly	To be measured	Pending
PT09	Concurrent Requests	Multiple users	Requests processed correctly	To be measured	Pending
PT10	End-to-End Flow	Complete patient request	Workflow completes successfully	To be measured	Pending

10. Performance Acceptance Criteria

Parameter	Suggested Target
Request Submission	Successful without transaction failure
Record Creation	Successful for valid requests
Diagnostic Test Processing	Correct test assignment and processing
Notification	Generated at configured stages
Error Rate	As low as practically achievable
Data Integrity	No missing or duplicated patient/diagnostic data
Concurrent Processing	No workflow or record corruption
Workflow Completion	Successful end-to-end execution
Scalability	Stable performance as workload increases

The exact numerical response-time and resource-utilization targets should be defined based on the actual ServiceNow environment and organizational requirements rather than assumed values.

11. Performance Testing Process

Identify Requirements → Define Test Scenarios → Prepare Environment → Prepare Test Data → Execute Tests → Monitor Metrics → Analyze Results → Identify Bottlenecks → Optimize → Retest → Document Results

12. Expected Outcome

Performance testing should demonstrate that the **Medical Diagnosis Centre Management System** can reliably handle patient requests, diagnostic test processing, record updates,

workflow execution, and notifications while maintaining acceptable responsiveness and data integrity.
It should also help identify potential performance bottlenecks and provide a basis for future optimization and scaling of the ServiceNow application.

